Sec: OSR.IIT_*COSC(MODEL-A&APEX) | GTA-3(P2) | Date: 05-04-26
Time: 3 Hrs | 2022_P2 | Max. Marks: 180

# KEY SHEET
## MATHS

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | **4** | 2 | **2** | 3 | **4** | 4 | **8** | 5 | **6** |
| 6 | **2** | 7 | **3** | 8 | **5** | 9 | **ABC** | 10 | **ACD** |
| 11 | **BD** | 12 | **AB** | 13 | **AB** | 14 | **AB** | 15 | **B** |
| 16 | **C** | 17 | **C** | 18 | **C** | | | | |

## PHYSICS

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 19 | **4** | 20 | **2** | 21 | **9** | 22 | **9** | 23 | **1** |
| 24 | **8** | 25 | **8** | 26 | **4** | 27 | **ABCD** | 28 | **AC** |
| 29 | **ACD** | 30 | **D** | 31 | **ACD** | 32 | **AC** | 33 | **B** |
| 34 | **D** | 35 | **A** | 36 | **D** | | | | |

## CHEMISTRY

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 37 | **9** | 38 | **6** | 39 | **4** | 40 | **5** | 41 | **2** |
| 42 | **3** | 43 | **6** | 44 | **2** | 45 | **ABCD** | 46 | **ABC** |
| 47 | **ABCD** | 48 | **BD** | 49 | **ACD** | 50 | **AB** | 51 | **D** |
| 52 | **B** | 53 | **C** | 54 | **D** | | | | |

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# SOLUTIONS

## MATHS

1. Given integral $=\frac{\pi}{2}\int_0^a \frac{1}{\tan^{-1}a+\tan^{-1}x}\cdot\frac{dx}{x^2+1}=\frac{\pi}{2}\left(\ln\left(\tan^{-1}a+\tan^{-1}x\right)\right)_0^a=\frac{\pi}{2}\ln 2$

$\therefore p+q=4$

2. $\because |z_2|=1\Rightarrow z_2=e^{i\theta}=\cos\theta+i\sin\theta$

$$\therefore \frac{z_2-1}{z_2+1}=\frac{\cos\theta-1+i\sin\theta}{\cos\theta+1+i\sin\theta}=\frac{-2\sin^2\frac{\theta}{2}+i\cdot 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2\cos^2\frac{\theta}{2}+i\cdot 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}=\frac{i\sin\frac{\theta}{2}}{\cos\frac{\theta}{2}}=i\tan\frac{\theta}{2}$$

$$\therefore \frac{z_1-1}{z_1+1}=-\tan^2\frac{\theta}{2}\Rightarrow z_1-1=-z_1\tan^2\frac{\theta}{2}-\tan^2\frac{\theta}{2}$$

$$\Rightarrow z_1=\frac{1-\tan^2\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}}=\cos\theta$$

Which is purely real and lies in $[-1,1]$

$\therefore |z_1-1|+|z_1+1|=2$

3. Let degree of $f(x)$ be $n$.

$\because$ degree of $f(f(f(x)))=$ degree of $f(x)$

$\Rightarrow \quad n^3=n \Rightarrow n=1$

$\therefore$ Let $f(x)=2x+b$ ($\because$ leading coefficient = 2)

$\therefore f(f(f(x)))=2(4x+3b)+b=8x+7b$

$\therefore 8+2-2p=0\Rightarrow p=5$

and $7b-4b=3\Rightarrow b=1$

$\therefore f(x)=2x+1$

$f(f(x))=4x+3$

$\therefore \frac{d}{dx}f(f(x))=4$

$f^{-1}(x)=\frac{x-1}{2}\Rightarrow 2f^{-1}(x)+1=x$

$\therefore \int_{-p}^{p}\left(2f^{-1}(x)+1\right)dx=0$

4. $\because$ Elements of determinant can be 1 or $-1$.

$\therefore$ First two places of first row can be filled in $2\times 2=4$ ways and last place can be filled in only one way to product be $-1$.

$\therefore$ First row can be made in 4 ways.

Similarly, second row can be made in 4 ways

And third row can be made 1 way only.

$\therefore$ Number of elements in $S=16=m$



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$\because \ |p| = \begin{vmatrix} 3 & -2 & 3 \\ 2 & -2 & 3 \\ 0 & -1 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 2 & -2 & 3 \\ 0 & -1 & 1 \end{vmatrix} = 1(-2+3) = 1$

$\because \ adj(adj\ P) = |P|^{n-2}.P = P$

$\Rightarrow tr.(adj(adj\,P)) = tr(P) = 2 = n$

$\therefore \dfrac{m}{n} = 8$

5. Equation of the line through $P(2, 3)$ is

$\dfrac{x-2}{\cos\theta} = \dfrac{y-3}{\sin\theta}(\theta = 30°)$

If $PA = r_1$, $PB = r_2$ then $r_1, r_2$ are the roots of the equation.

$(2 + r\cos\theta)^2 - 2(2 + r\cos\theta)(3 + r\sin\theta) - (3 + r\sin\theta)^2 = 0$

$\Rightarrow r^2(\cos 2\theta - \sin 2\theta) - 2r(\cos\theta + 5\sin\theta) - 17 = 0$

So $a = PA \cdot PB = r_1 r_2 = \dfrac{-17}{\cos 2\theta - \sin 2\theta} = 17(\sqrt{3}+1)$

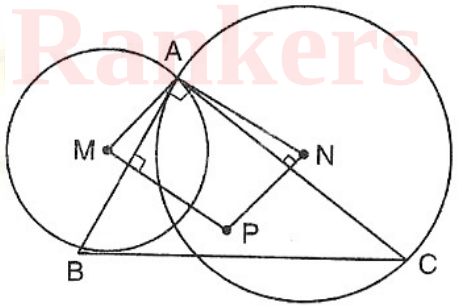
6. Clearly PM is perpendicular bisector of AB and PN is perpendicular bisector of AC

$\therefore$ P is circumcentre of $\Delta ABC$

$\therefore \ PA = PB = PC = R$ (circumradius)

$\therefore \ a + b + c = 2R\sin A + 2R\sin B + 2R\sin C$

$\therefore \ K = 2$




7. Let $\vec{m} = a\hat{i} + b\hat{j} + c\hat{k}$

and $\vec{n} = p\hat{i} + q\hat{i} + r\hat{k}$

$\therefore \quad \vec{m} \cdot \vec{n} = -|\vec{m}||\vec{n}|$

$\therefore \quad \vec{m}$ is antiparallel to $\vec{n}$

$\therefore \quad \dfrac{a}{p} = \dfrac{b}{q} = \dfrac{c}{r}$

$\therefore \quad \dfrac{aq}{bp} + \dfrac{br}{cq} + \dfrac{cp}{ar} = 3$

8. Let the equation of the circle be $x^2 + y^2 + 2gx + 2fy + c = 0$. Since it passes through $(3, 4)$, $6g + 8f + c = -25$. As it cut the circle $x^2 + y^2 = a^2$ orthogonally

$\Rightarrow 6g + 8f + a^2 + 25 = 0$

Locus of the centre $(-g, -f)$ is $6x + 8y - (a^2 + 25) = 0$



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Distance of the line from the origin is $3=\dfrac{a^2+25}{\sqrt{36+64}}\Rightarrow a^2=5$

9. Let $y=mx+c$ be a chord of the given curve. Equation of the pair of lines through the origin and the points of intersection of the chords and the curve is

$$3x^2-y^2-(2x-4y)\left(\frac{y-mx}{c}\right)=0$$

If these are at right angles

$$3+\frac{2m}{c}+\left(-1+\frac{4}{c}\right)=0$$

$\Rightarrow m=-(c+2)$.

So the equation of the chord is

$y+(c+2)x-c=0\Rightarrow (y+2x)+c(x-1)=0$

Showing that the chord passes through the point of intersection of the lines $y+2x=0$ and $x-1=0\Rightarrow$ through the point $(1,-2)$ and the equation of the parabola in (c) is

$(x-1)^2=4(y+2)$ whose vertex is also $(1,-2)$.

**Note:** The centre of the circle in (d) is $(-1,2)$.

Equation of the rectangular hyperbola in

(a) can be written as $(x-1)^2-(y+2)^2=9$ centre of which is $(1,-2)$

10. Given $P(A\cup B\cup C)=\dfrac{3}{4}$ …(1)

$$P(A\cap B\cap\bar{C})+P(A\cap\bar{B}\cap C)+P(\bar{A}\cap B\cap C)+P(A\cap B\cap C)=\frac{1}{2}\quad\ldots(2)$$

And $P(A\cap B\cap\bar{C})+P(A\cap\bar{B}\cap C)+P(\bar{A}\cap B\cap C)=\dfrac{2}{5}$ …(3)

$\therefore$ From equation (2) $-$ equation (3)

$P(A\cap B\cap C)=\dfrac{1}{2}-\dfrac{2}{5}=\dfrac{1}{10}=P(ABC)\Rightarrow$ (A) is correct

$\therefore P(A\cap B)+P(B\cap C)+P(C\cap A)=$

$$P(A\cap B\cap\bar{C})+P(A\cap\bar{B}\cap C)+P(\bar{A}\cap B\cap C)+3P(A\cap B\cap C)$$

$=\dfrac{2}{5}+3\times\dfrac{1}{10}=\dfrac{7}{10}\Rightarrow$ (B) is incorrect

$\therefore P(A)+P(B)+P(C)=P(A\cup B\cup C)+P(AB)+P(BC)+P(CA)-P(ABC)$

$=\dfrac{3}{4}+\dfrac{7}{10}-\dfrac{1}{10}=\dfrac{27}{20}\Rightarrow$ (C) is correct

Also $P(A\cap\bar{B}\cap\bar{C})+P(\bar{A}\cap\bar{B}\cap C)+P(\bar{A}\cap B\cap\bar{C})=P(A)+P(B)+P(C)$

$$-2\left[P(A\cap B)+P(B\cap C)+P(C\cap A)\right]+3P(A\cap B\cap C)$$

$=\dfrac{27}{20}-\dfrac{14}{10}+\dfrac{3}{10}=\dfrac{5}{20}=\dfrac{1}{4}\Rightarrow$ (D) is correct



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11. $\vec{d}\cdot\vec{a}=\begin{bmatrix}a & b & c\end{bmatrix}\cos y$

$\vec{d}\cdot\vec{c}=\begin{bmatrix}a & b & c\end{bmatrix}\sin x$

$\vec{d}\cdot\vec{b}=2\begin{bmatrix}c & a & b\end{bmatrix}=2\begin{bmatrix}a & b & c\end{bmatrix}$

$\vec{d}\cdot\left(\vec{a}+\vec{b}+\vec{c}\right)=0\Rightarrow\cos y+\sin x+2=0$

$\Rightarrow\cos y=-1,\ \sin x=-1$

$\Rightarrow x^2+y^2\geq\left(\frac{\pi}{2}\right)^2+\pi^2=\frac{5\pi^2}{4}$

12. Given $D.E^n$ is $\frac{dy}{2x\,dx}+\frac{y}{x^2}=\frac{\sin^{-1}x}{2x^2}$

$\Rightarrow\frac{dy}{dx}+\left(\frac{2}{x}\right)y=\frac{\sin^{-1}x}{x}$

I.F. $=e^{\int\frac{2}{x}dx}=e^{2\ln x}=x^2$

Solution is given by $y\times x^2=\int\frac{\sin^{-1}x}{x}\times x^2dx$

$\Rightarrow y\times x^2=\frac{1}{4}\left(2x^2-1\right)\sin^{-1}x+\frac{1}{4}x\sqrt{1-x^2}+C$ at $x=0,\ y=0\Rightarrow c=0$

13. $\mathrm{A}=\cos\theta.\cos2\theta.\cos3\theta...\cos\left(n-1\right)\theta$

$\mathrm{B}=\sin\theta.\sin2\theta.\sin3\theta...\sin\left(n-1\right)\theta$

Where $\theta=\frac{2\pi}{2019}$

$n=2019$

$2^{1009}\times\mathrm{A}\times\mathrm{B}=\left(2\sin\theta.\cos\theta\right)\left(2\sin2\theta.\cos2\theta\right)....\left(2\sin\left(n-1\right)\theta.\cos\left(n-1\right)\theta\right)$

$=\sin2\theta.\sin4\theta\ \sin6\theta....\sin\left(2n-2\right)\theta$

$\Rightarrow2^{1009}\times\mathrm{A}\times\mathrm{B}=\mathrm{B}$

$\mathrm{A}=\frac{1}{2^{1009}}=2^{-1009}$

$\frac{p}{q}=\frac{1009}{1}\Rightarrow p+q=1010$

14. $I_n=\int_0^\infty x^ne^{-x^2}dx=\int_0^\infty x^{n-1}xe^{-x^2}dx$

$\Rightarrow I_n=\left[x^{n-1}\int xe^{-x^2}dx\right]_0^\infty-\int_0^\infty\left\{\frac{d}{dx}\left(x^{n-1}\right)\int xe^{-x^2}\right\}dx$

$-x^2=t\Rightarrow I_n=-\left[\frac{x^{n-1}}{2e^{x^2}}\right]_0^\infty+\frac{n-1}{2}\int_0^\infty x^{n-2}e^{-x^2}dx$

$\Rightarrow I_n=\frac{n-1}{2}I_{n-2}$



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Case I: $n=2m \Rightarrow I_n=\dfrac{(2m-1)(2m-3)(2m-5).....1}{2^m}I_0=\dfrac{(2m-1)!}{2^{2m-1}(m-1)!}a$

15. LHS $=\int_0^{4\pi}\ln\left|14\sin(x+\alpha)\right|dx$ where $\cos\alpha=\dfrac{13}{14}$

$=4\left(\int_0^{\pi}\ln\left|\sin(x+\alpha)\right|dx+\pi\ln 14\right)$

Let $x+\alpha=t$'

$=4\int_{\alpha}^{\pi+\alpha}\ln\sin t\ dt+4\pi\ln 14$

$=4\pi\ln 14+4\times 2\times\left(\dfrac{-\pi}{2}\ln 2\right)=4\pi\ln 7$

16. Equation of tangent on $\dfrac{x^2}{3}+\dfrac{y^2}{2}=1$ will be

$y=mx\pm\sqrt{3m^2+2}$

$\because$ Equation (i) coincide with $y=kx+2h$

$\therefore\ m=k\ \&\ \sqrt{3m^2+2}=2h$

$\Rightarrow 3k^2+2=4h^2\Rightarrow 4h^2-3k^2=2$

$\therefore$ locus of $(h,k)$ is

$\dfrac{x^2}{2/4}-\dfrac{y^2}{2/3}=1$

$\therefore$ eccentricity $=\sqrt{1+\dfrac{\frac{2}{3}}{\frac{2}{4}}}=-\sqrt{1+\dfrac{4}{3}}=\sqrt{\dfrac{7}{3}}$

17. $\because\ z_1$ satisfies $\arg z=\dfrac{\pi}{4}$

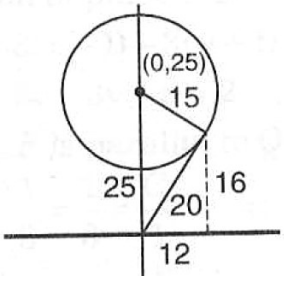
$\therefore\ z_1=\alpha+\alpha i,\quad \alpha>0$

$\because\ \left|z_1-2-i\right|=1\Rightarrow\left|\alpha+\alpha i-2-i\right|=1$

$\Rightarrow(\alpha-2)^2+(\alpha-1)^2=1\Rightarrow\alpha^2-3\alpha+2=0\Rightarrow\alpha=1$ or $2$

$\therefore\ z_1=1+i$ or $2+2i$

For $z_2$




Clearly $z_2=12+16i$

$\therefore$ min value of $\left|z_1-z_2\right|=\left|10+14i\right|=\sqrt{296}=2\sqrt{74}$



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18. $\because$ LCM of $\alpha, \beta, \gamma = p^3q^2r$ & HCF $= pqr$

$\therefore \quad \alpha = p^{m_1}q^{n_1}r$

$\beta = p^{m_2}q^{n_2}r$

$\gamma = p^{m_3}q^{n_3}r$

Minimum of $(m_1, m_2, m_3) = 1$ & Maximum of $(m_1, m_2, m_3) = 3$

$\therefore$ Number of possibilities for $m_1, m_2, m_3 = 12$

and minimum of $(n_1, n_2, n_3) = 1$ and maximum $(n_1, n_2, n_3) = 2$

$\therefore$ Number of possibilities = 6

$\therefore$ Total number of ordered triplets $= 12 \times 6 = 72$

## PHYSICS

19. $E(t) = A^2e^{-\alpha t}$

$ln\ E = 2\ln\ A - \alpha t \therefore \frac{\Delta E}{E} \times 100 = 2\frac{\Delta A}{A} \times 100 - \alpha\Delta t \times 100$

$\frac{\Delta t}{t} \times 100 = 1.5$, $At\ t = 5s.\Delta t = \frac{1.5 \times 5}{100} = \frac{7.5}{100}$

% error in $E = 2 \times (1.25) + \alpha \times \frac{7.5}{100} \times 100 = 2.5 + 1.5 = 4$

20. Taking torque about point A

$$N\sqrt{\ell^2 - (\ell - h)^2} = f_s(\ell - h)$$

$$\Rightarrow f_s = \frac{N\sqrt{\ell^2 - (\ell - h)^2}}{\ell - h} \le \mu N$$

$$\Rightarrow \ell^2 - (\ell - h)^2 \le \mu^2(\ell - h)^2$$

$$\Rightarrow \ell^2 \le (\mu^2 + 1)(\ell - h)^2$$

$$\Rightarrow \frac{\ell}{\sqrt{\mu^2 + 1}} \le (\ell - h)$$

$$\Rightarrow h \le \ell\left[1 - \frac{1}{\sqrt{1 + \mu^2}}\right] \Rightarrow h_{max} = 2\ m$$

$f_s = 3\ N$

$N = 4\ N$

$F_H = 5\ N$

21. $\cos\theta = \frac{\frac{(\sqrt{5}+1)}{4}R}{R}$

$= \frac{\sqrt{5}+1}{4}$

$\therefore \theta = 36^\circ$



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Clearly $\phi = 2\times 54^\circ = 108^\circ$

Ray will be incident at A when

$n\phi = m\times 360^\circ$ [m, n are integers]

Obviously, $m = 3, n = 10$

∴ Light ray will suffer 9 reflections before reaching A again.

22. C.M. of the cube comes down through a height of $\sqrt{3}$ m while C.M. Of displaced water goes up by a height of $\frac{\sqrt{3}}{2}$ m.

⇒ Loss in PE of cube $= 2000\times 10\times\sqrt{3}$ Joule.

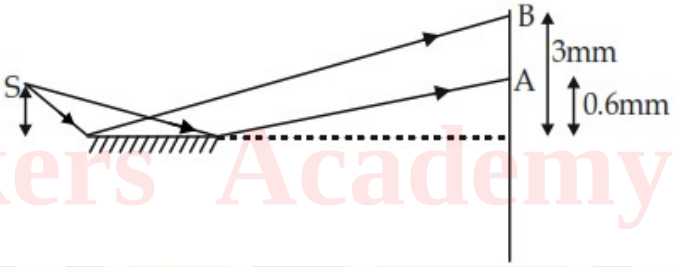
$= 2\sqrt{3}\times 10^4$ Joule.

Gain in PE of water $= 1000\times 10\times\frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2}\times 10^4$ Joule

23. At A $\frac{(0.6)(1.2)\times 10^{-6}}{6} = n_1\left(4\times 10^{-7}\right)$

$\Rightarrow n_1 = 0.3$

At B $\frac{(3)(1.2)\times 10^{-6}}{6} = n_2\left(4\times 10^{-7}\right) \Rightarrow n_2 = 1.5$




So, there will be one minima in the pattern.

24. Flux through curved surface is total flux minus the flux through the two circular faces.

25. For parallel process $t_{1/2} = \frac{1}{10}hr$

Total number of half lives $= \frac{1/2}{1/4} + \frac{1}{1/6} + \frac{1/2}{1/10}$ ∴ $n + 5 = 13 = 13$ $n = 8$

26. At $t = T/6$, particle will travel only $1/6$ of the circle.

Average speed

$= \frac{\text{ArcAB}}{\text{Time}} = \frac{(2\pi R)/6}{T/6} = \frac{2\pi R}{T}$

Average velocity

$= \frac{\text{Chord AB}}{\text{Time}} = \frac{2R\sin 30^\circ}{T/6} = \frac{6R}{T}$

Distance $= \frac{R}{T}\left[\frac{44}{7} - 6\right] = \frac{2R}{7T} = 2$ ms$^{-1}$ (given)

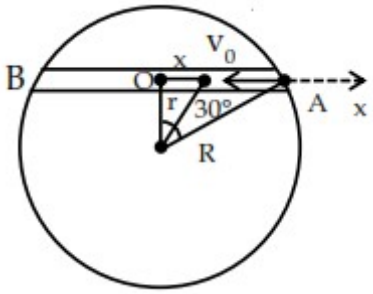
$R = 7$ $T = 7(2.3) = 16.10$ m

27. The system spins in the CM frame at a constant angular speed.



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28.




$$\frac{d^2x}{dt^2} = -\frac{GMr\cos\theta}{R^3} = -\left(\frac{GM}{R^3}\right)x = -\left(\frac{g_0}{R}\right)x$$

$$\omega = \sqrt{\frac{g_0}{R}}$$

Now, $x = A\sin(\omega t + \alpha)$ at $t = 0, x = +R/2$

$$\frac{R}{2} = A\sin\alpha$$

$V_x = \omega A\cos(\omega t + \alpha)$ at $t = 0, v_x = -\frac{\sqrt{3g_0R}}{2}$

$$-\frac{\sqrt{3g_0R}}{2} = \sqrt{\frac{g_0}{R}}A\cos\alpha$$

$$-\frac{\sqrt{3}}{2}R = A\cos\alpha$$

Solving (i) and (ii), we get

$$A = R, \alpha = \frac{5\pi}{6}$$

$$x = R\sin\left(\sqrt{\frac{g_0}{R}}t + \frac{5\pi}{6}\right)$$

$$-\frac{R}{2} = R\sin\left(\sqrt{\frac{g_0}{R}}t + \frac{5\pi}{6}\right)$$

$$\frac{7\pi}{6} = \sqrt{\frac{g_0}{R}}t + \frac{5\pi}{6}$$

$$t = \frac{\pi}{3}\sqrt{\frac{R}{g_0}}$$

$$v_{max} = \omega A$$

$$v_{max} = \sqrt{\frac{g_0}{R}}R = \sqrt{g_0R}$$

29. For isothermal process : $Q = W \Rightarrow tan\theta = 1$

For adiabatic process : $Q = 0 \Rightarrow \frac{W}{Q} = \infty \Rightarrow tan\theta = \infty$

For isobaric process : $Q = nC_p\Delta T$ and



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$$W = n\left(C_P - C_V\right)\Delta T$$

$$\Rightarrow tan\theta = \frac{W}{Q} = \frac{C_P - C_V}{C_P} = 1 - \frac{C_V}{C_P} = 1 - \frac{1}{\gamma}$$

30. $\vec{B}\cdot\vec{d\ell} = \frac{\mu_0 i}{2\pi}\Delta\theta$, where $\Delta\theta$ is angle subtended at wire by ends of path

31. A rectangular toroidal coil of height $h$, inner radius $a$, outer radius $b$, number of turns $N$, and current I in each turn.
Magnetic field at radial distance $r$ from the center $(a < r < b)$ :
Using Ampère's law:

$$\oint \vec{B}\cdot d\vec{\ell} = \mu_0 I_{encl} \Rightarrow B(r)\cdot 2\pi r = \mu_0 NI \Rightarrow B(r) = \frac{\mu_0 NI}{2\pi r}$$ This option is correct.

Total magnetic flux linking the coil:
The magnetic flux through a differential strip of height $h$ and thickness dr at radius $r$ is:

$$d\Phi = B(r)\cdot h\cdot dr = \frac{\mu_0 NI}{2\pi r}hdr$$

Total flux linked through N turns:

$$\Phi = N\int_a^b \frac{\mu_0 NIh}{2\pi r}dr = \frac{\mu_0 N^2 Ih}{2\pi}\int_{ar}^{b}\frac{1}{r}dr = \frac{\mu_0 N^2 Ih}{2\pi}\ln\left(\frac{b}{a}\right)$$

32. $Z_1 = R_1 + i\omega L$, $Z_2 = R_2$

$$Z_3 = R_3 + \frac{1}{i\omega C_3},\ Z_4 = \frac{1}{i\omega C_4}$$

For bridge to be balanced

$$Z_1 Z_4 = Z_2 Z_3$$

$$\frac{R_1}{i\omega C_4} + \frac{L_1}{C_4} = R_2 R_3 + \frac{R_2}{i\omega C_3}$$

$$\Rightarrow \frac{L_1}{C_4} = R_2 R_3;\ \frac{R_1}{C_4} = \frac{R_2}{C_3}$$

33. Once the ball is fully unraveled (i.e. there is only a straight string), its centre of mass is exactly in one half of its length, which is at height $\frac{\ell}{2}\sin\alpha$. In the beginning, the centre of mass of the ball is at height $R\cos\alpha$. Therefore, the difference in potential energy is $\Delta E = Mg\left(\frac{\ell}{2}\sin\alpha - R\cos\alpha\right)$. Let v denote the velocity of the kick. The initial kinetic energy of translation is $\frac{1}{2}Mv^2$ and the kinetic energy of rotation is $\frac{1}{2}I\frac{v^2}{R^2}$, where $I = \frac{2}{5}MR^2$.

Therefore, we obtain $\Delta E = \frac{1}{2}Mv^2 + \frac{1}{5}Mv^2 = \frac{7}{10}Mv^2$

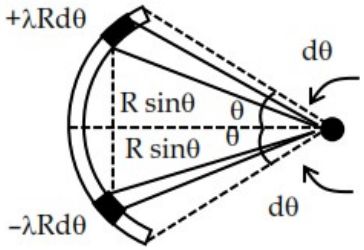
$$Mg\left(\frac{\ell}{2}\sin\alpha - R\cos\alpha\right) = \frac{7}{10}Mv^2$$

$$v = \sqrt{\frac{5g}{7}\left(\ell\sin\alpha - 2R\cos\alpha\right)} \doteq 6.06\ m\cdot s^{-1}$$



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34. We consider two differential elements on the rod as shown in figure. These elements constitute an electric dipole whose dipole moment is $|dP| = (\lambda R d\theta)(2R\sin\theta)$




Net dipole moment of the system is $= 2\lambda R^2(1-\cos\theta_1) = 4\lambda R^2\sin^2(\theta_0/2)$

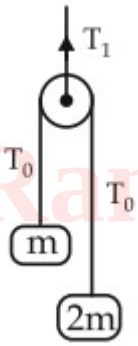
Time period of oscillation of a dipole in a uniform magnetic field is

$$T = 2\pi\sqrt{\frac{I}{pE}}$$

$$= 2\pi\sqrt{\frac{mR^2}{4\lambda R^2\sin^2(\theta_0/2)E}}$$

$$= \pi\sqrt{\frac{m}{\lambda\sin^2(\theta_0/2)E}}$$

35.




$$T_1 = 2\ T_0 = 2\left[\frac{2\,m(2\,m)}{m+2\,m}\right]g$$

$$T_1 = \frac{8\,m}{3}\,g = \frac{80\,m}{3}\ldots\ldots\ldots$$

In resonance,

$$f_{wire} = f_{tube}$$

$$\frac{(1)V_1}{2t_1} = \frac{(1)V_2}{4t_2}$$

$$\frac{\left(\sqrt{\frac{T_1}{n}}\right)}{2(x)} = \frac{(400)}{4\left(\frac{3}{2}\right)}$$

$$\Rightarrow T_1 = m(16\times10^4)$$

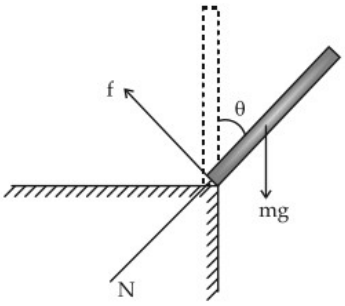
$$\frac{80}{3}\,m = 10^{-4}(16\times10^4)$$

$$m = 0.6\ kg$$



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36.




$$\frac{1}{2}\frac{m\ell^2}{3}\omega^2=\frac{mg\ell}{2}(1-\cos\theta)$$

$$mg\cos\theta-N=\frac{m\omega^2\epsilon}{2}$$

$$N=\frac{mg}{2}(5\cos\theta-3)$$

N becomes zero when $\cos\theta=\frac{3}{5}$

$$\therefore \sin\theta=\frac{4}{5}$$

## CHEMISTRY

37. Oxide (A) $M_2O_x$.

$$\frac{2}{x}=\frac{w(m)}{A(m)}:\frac{w(o)}{A(o)}$$

$$\frac{2}{x}=\frac{0.7745}{A(m)}:\frac{0.2255}{16}$$

$$\frac{2}{x}=\frac{0.7745\times16}{0.2255\times A(m)}=\frac{54.95}{A(m)}$$

Oxide (B) $M_2O_y$

$$\frac{2}{y}=\frac{w(m)}{A(m)}:\frac{w(o)}{A(o)}$$

$$\frac{2}{y}=\frac{0.4952}{A(m)}:\frac{0.5048}{16}=\frac{15.695}{A(m)}$$

So that $\frac{y}{x}=\frac{54.95}{15.695}=3.5$

So that $\frac{y}{x}=\frac{7}{2}$

$A(m)=54.95$

Metal $M$ is $Mn$

Oxide A $\rightarrow MnO$ (green colour crystal)

Oxide B $\rightarrow$ $Mn_2O_7$ (dark green)

$\therefore x+y=9$



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38. $\left[Cr(H_2O)_6\right]Cl_3 Cr^{3+}\ 3d^3\ 3$ unpaired electrons

$\left[Cr(en)_3\right]Cl_3\ Cr^{3+}\ 3d^3\ 3$ unpaired electrons

$\left[Cr(NH_3)_4Cl_2\right]Cl\ Cr^{3+} 3d^3\ 3$ unpaired electrons

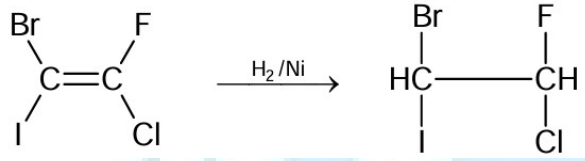
$\left[Cr(H_2O)_5Cl\right]SO_4\ Cr^{3+} 3d^3\ 3$ unpaired electrons

$\left[Co(H_2O)_4Cl_2\right]NO_3\ Co^{3+}\ 4$ unpaired electrons

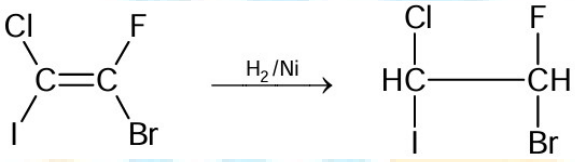
$K_3\left[Fe(CN)_6\right]\ Fe^{+3}$ 1 unpaired electron

39. $BF_3, SbCl_3, SiF_4, SF_6$

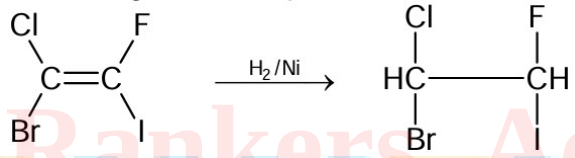
40.




E (or) z give same product.

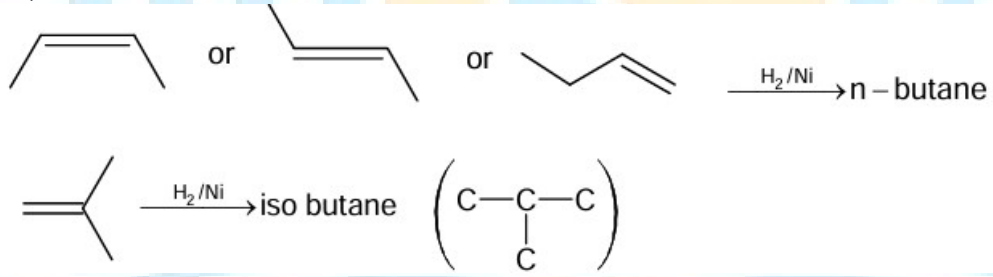



E (or) z give same product.




E (or) z give same product.

So, $x = 3$




$Y = 2$

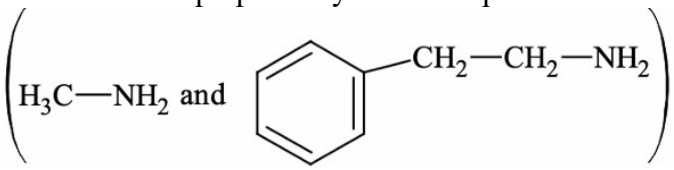
$\therefore\ X + Y = 5$

41. One mole of the above compound (X) on hydrolysis followed by heating produces 4 moles of $CO_2$.

Hence, 0.5 moles of X produce 2 moles of $CO_2$.

42. Stronger base than benzyl amine (a) = 5

Amines can be prepared by Gabriel's phthalimide method (b) = 2




43. $B_2, C_2, N_2, O_2, F_2, CO$ are correct.

All even $e^-$ species will become diamagnetic in such condition.



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44. $\Delta T_f = iK_f m, \qquad m = \frac{20}{172} \times \frac{1000}{50} = \frac{400}{172}$

$$i = \frac{\Delta T_f}{K_f \times m} = \frac{2}{1.72 \times \frac{400}{172}}$$

$i = 0.5$

45. (A) Radial wave function contains information about $n$ and $l$

(B) $R_{(2,0)}^{(r)} = \frac{1}{\sqrt{8}}\left[\frac{1}{a_0}\right]^{3/2}\left(2 - \frac{r}{a_0}\right)e^{-r/2a_0}$

46. (A) $-\frac{d[A]}{dt} = \frac{d[B]}{dt} + \frac{d[C]}{dt}$

(B) $k[A] = k_1[A] + k_2[A]$

Where $k$ is overall rate constant.

$k = k_1 + k_2$

Or $\frac{dk}{dT} = \frac{dk_1}{dT} + \frac{dk_2}{dT}$

Or $Ae^{-\frac{Ea}{RT}}\left[\frac{Ea}{RT^2}\right] = A_1e^{-\frac{Ea_1}{RT}}\left[\frac{Ea_1}{RT^2}\right] + A_2e^{-\frac{Ea}{RT}}\left[\frac{Ea}{RT^2}\right]$

Or $k.Ea = k_1Ea_1 + k_2Ea_2$

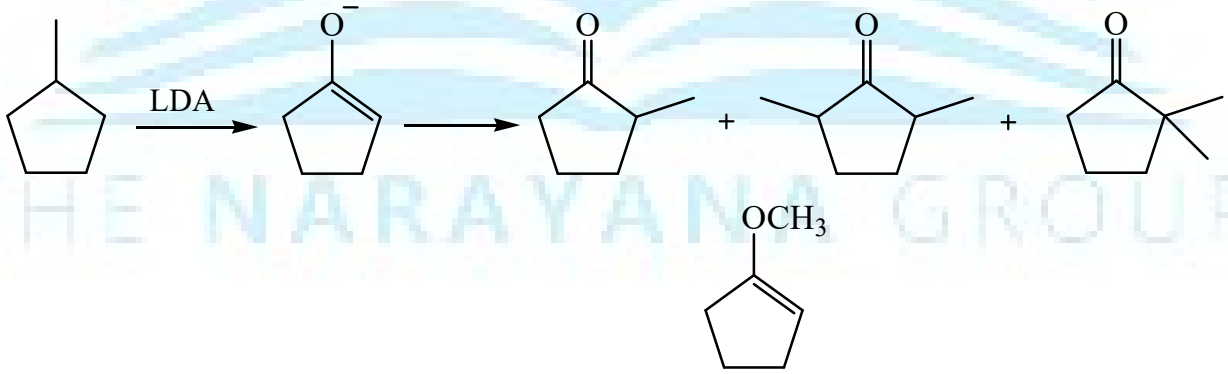
Or $Ea = \frac{k_1E_{a_1} + k_2E_{a_2}}{k} = \frac{x_1y_1 + x_2y_2}{x_1 + y_2}$

(C) $t_{1/2} = \frac{\ln 2}{k} = \frac{\ln 2}{x_1 + x_2}$

(D) $\ln\frac{k'}{k} = \frac{E_a}{R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$

Higher the activation energy of a chemical reaction, more pronounced effect of temperature.

47.




48. Pine oil is frother and xanthates are collectors.

49. Work is a form of exchange energy between system and surrounding in adiabatic process.
- Intensive property is not additive.
- In free expansion work done is zero.
- For an isolated system the entropy either increases or remains constant.

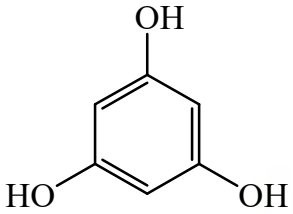
50. Fact based question.



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51. $Bi^{3+} + 3I^- \rightarrow \underset{\text{Black}}{BiI_3} \xrightarrow{H_2O} \underset{\text{Orange ppt}}{BiOI}$

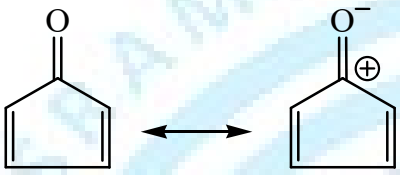
52. $Cl^-$ ion constituting the $KCl$ in the salt bridge migrates towards the half cell in which oxidation occurs and hence this half-cell should not have any ion reversible w.r.t to electrode which may precipitate by $Cl^-$ ion or the half - cell must not be reversible w.r.t $Cl^-$ ion.

53. 'b' exist as 

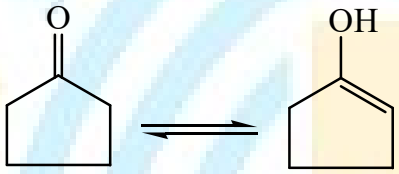

In 'a' due to steric factor, very less enolization occurs.

In e : $C-O$ bond order is 1.33

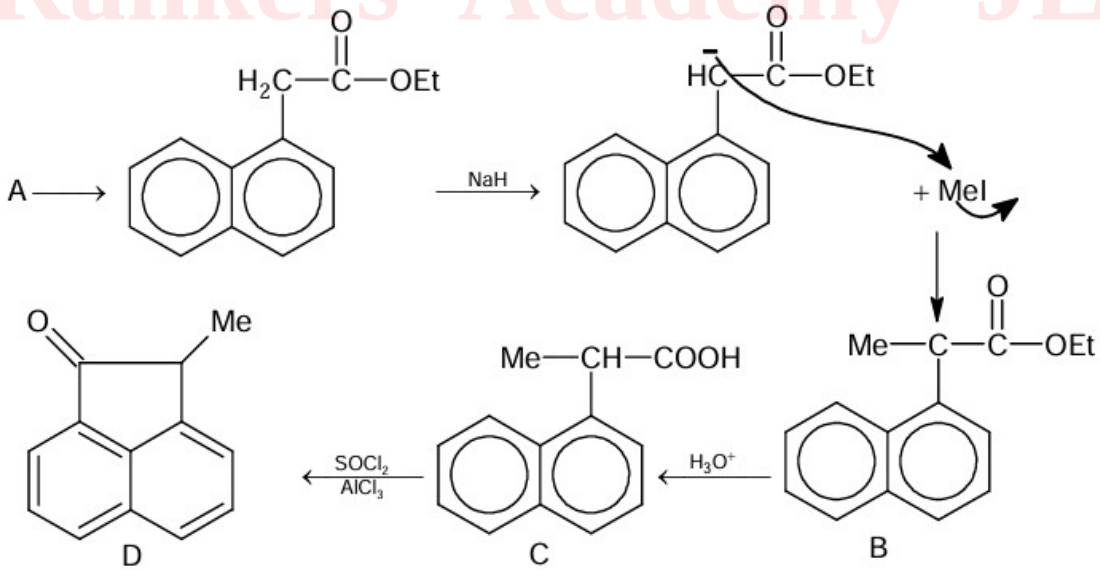
In f : $C-O$ bond order is 1.5

In c : 

(anti aromatic)

In d : 

(Major) (Minor)

54. 




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